

Question				Marking details	Marks Available																	
					AO1	AO2	AO3	Total	Maths	Prac												
4	(a)	(i)		Molecules with the same {molecular / chemical} formula, but different {arrangement of atoms / structure / structural formula} (1)	1			1														
		(ii)		<table><tr><td>A</td><td>B</td></tr><tr><td>starch/ amylose/ amylopectin</td><td>cellulose</td></tr><tr><td>α- glucose</td><td>β – glucose</td></tr><tr><td colspan="2">All four boxes above correct = 2 marks 2/3 boxes correct = 1 mark 0/1 box correct = 0 marks</td></tr><tr><td>1-4 glycosidic bonds / {helical / spiral / compact} structure (1) Ignore reference to branching</td><td>each {monomer / glucose} flipped by 180° (with respect to previous one) / straight chains / reference to {hydrogen bonds between adjacent chains / microfibrils} / 1-4 glycosidic bonds (1)</td></tr><tr><td>{Carbohydrate / glucose} store (1)</td><td>cell wall / structural / rigidity / strength (1)</td></tr></table>	A	B	starch/ amylose/ amylopectin	cellulose	α- glucose	β – glucose	All four boxes above correct = 2 marks 2/3 boxes correct = 1 mark 0/1 box correct = 0 marks		1-4 glycosidic bonds / {helical / spiral / compact} structure (1) Ignore reference to branching	each {monomer / glucose} flipped by 180° (with respect to previous one) / straight chains / reference to {hydrogen bonds between adjacent chains / microfibrils} / 1-4 glycosidic bonds (1)	{Carbohydrate / glucose} store (1)	cell wall / structural / rigidity / strength (1)	4	2		6		
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{Carbohydrate / glucose} store (1)	cell wall / structural / rigidity / strength (1)																					
		(iii)		Hydrolysis is {chemical insertion / addition} of water to break the bond (1) Maltose (1)	2			2														

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	(b)			Any one (x1) from - Both have a glycosidic bond (1) - Made of (two) glucose (monomers) (1) Any two (x1) from - In {maltose / 4.3} monomers are joined by an (α-)1,4 (glycosidic bond), in {gentiobiose / 4.4} they are joined by a (β-)1,6 (glycosidic bond). {maltose / 4.3} – α glucose, {gentiobiose / 4.4} β glucose {Maltose / 4.3} – OH group on carbon 1 points down and {gentiobiose / 4.4} β glucose OH group on carbon 1 points up / owtte		3		3		
	(c)	(i)		Correct annotation showing oxygen as $\delta -$ and hydrogen as $\delta +$ on any of the water molecules (1) NOT - /+	1			1		
		(ii)		{O-H / hydroxyl / polar} (groups) will form <u>hydrogen</u> bonds with water (molecules) (1)			1	1		
		(iii)		for transport / or description of (1) between {cells / tissues/ named tissues e.g. phloem} / through plasmodesmata (1)	2			2		
				Question 4 total	10	5	1	16		